

# Kai-Ju Yang

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## >Summary

Senior Software Engineer with 9+ years of experience in semiconductor automation, specializing in scalable system architecture, backend development, and hardware-software integration. Proven ability to design high-performance, real-time systems and automation frameworks using Python, C#, and C++. Strong background in distributed systems, communication protocols (SECS/GEM, TCP/IP), and data-driven optimization, with hands-on experience building simulation platforms, validation tools, and full-stack applications (React, FastAPI, Django). Recently expanded into AI and machine learning, developing deep learning-based defect detection systems and data analytics pipelines. Passionate about building robust backend systems and intelligent applications that bridge real-world systems with scalable software.

## >SKILLS

- Core Languages: Python (Automation, Data Parsing, NumPy/Pandas), C# (WPF, Desktop Tools), C/C++ (Embedded, High-performance Modules), MATLAB.
- Automation & Control: Semiconductor Equipment Automation (EFEM), SECS/GEM Protocols, TCP/IP & Serial Communication (UART/SPI/I2C), Multi-threaded Programming.
- Validation & Testing: Automated Test Framework Design, Log Analysis & Root Cause Analysis, System Simulation (Digital Twin), Unit Testing.
- Tools & Platforms: Linux (Ubuntu/CentOS), Git, CI/CD, Docker, STM32CubeIDE, OpenCV, PyTorch (AI-based Defect Detection)
- Domain Knowledge: Process Control, Yield Analysis, Signal Processing (FIR/IIR Filters), Control Theory.

## >EDUCATION

**Master of Automation and Precision Engineering,** *graduated at NYCU 2022.09 – 2025.06*

- Thesis: Advanced Defect Detection using Deep Learning Architectures.
- Relevant Coursework: Digital Signal Processing (DSP), Control System Design (FIR/IIR Filters in MATLAB), Microprocessor System Design.

**Master of Semiconductor Material and Process,** *studied at NCTU 2020.9 - 2022.01*

**Bachelor of Chemical and Material Engineer,** *graduated at Yuntech, 2011.09-2015.06*

## >PROFESSIONAL EXPERIENCE

**SunSun Technology Co., LTD. | Assistant Manager | Taiwan** **2025.11 ~ 2026.03**

*Led a cross-functional software team to develop and integrate hardware/software systems for high-precision fiber array bonding equipment in Silicon Photonics applications.*

- Designed a high-precision automatic alignment control system by integrating four-view CCD camera vision algorithms. Implemented real-time computer vision compensation and multi-axis motor control, achieving ultra-high alignment accuracy at the 2  $\mu\text{m}$  level.
- Architected equipment communication frameworks and data exchange specifications. Developed low-level communication handlers to coordinate data flow and command logic between machines, vendors, and host systems.
- Drove equipment standardization and smart manufacturing integration by ensuring compliance with semiconductor automation standards (SECS/GEM), enabling seamless interoperability with factory systems (CIM/EAP).
- Initiated and developed in-house process data visualization and SPC software.
- Built real-time analytics dashboards to monitor production trends, improving customer satisfaction while significantly reducing

**Jcc Software Technology Co., LTD. | Assistant Manager | Taiwan**

**2021.06 ~ 2025.11**

*Led the software architecture and system integration for high-precision semiconductor automation equipment, focusing on reliability and validation efficiency.*

- Automation Framework Architecture: Architected the software core for a new EFEM (Equipment Front End Module) product line, integrating complex hardware logic with upper-level CIM systems. Resulted in stable annual sales of 40-60 units and generated ~\$1.2M-\$1.8M in revenue
- Virtual Validation Tool : Developed a proprietary Simulation Testing Tool to emulate hardware behaviors and communication protocols. This tool allowed for software pre-validation without physical machines, reducing on-site debugging time and costs by 33% and significantly improving software quality before deployment.
- Legacy Optimization & Cost Reduction: Refactored legacy systems by developing high-performance C++ modules to replace expensive third-party packages, reducing licensing fees and increasing project profit margins by 4% while enhancing system stability.
- Cross-Functional Leadership: Managed a team of developers and coordinated with hardware vendors to define specifications for equipment integration, ensuring 100% compliance with fabrication plant requirements.

#### **Tokyo Electron Ltd.| Process Engineer | Taiwan**

2018.02 - 2021.05

- Process Optimization: Developed process recipes that reduced cycle time by 11% while improving profile health by 9%, demonstrating a deep understanding of semiconductor physics and manufacturing constraints.
- Data Automation: Developed VB macros to automate the consolidation and analysis of massive raw data sets, increasing engineering data processing efficiency by 60%.

#### **Premtek| Process Engineer | Taiwan**

2016.01 - 2018.01

- Yield Enhancement: Achieved a 13% reduction in defect rates and a 20% yield increase through statistical parameter analysis and systematic process tuning.

### **>PROJECT HIGHLIGHTS**

#### **SECS/GEM Communication Validation Tool (C#, WPF, TCP/SECS) 2025.09 ~ 2025.11**

*Used by QA/FAE to validate SECS/GEM communication without real hardware, enabling issue reproduction and root cause debugging in software-only environment.*

- Developed a configurable EAP Host Emulator to simulate complex semiconductor communication scenarios.
- Implemented automated verification flows and dynamic parameter controls, enabling engineers to reproduce field issues in a software-only environment, drastically shortening the debugging cycle.
- Relevance: Demonstrates capability in building Protocol Validation Tools.
- Online: <https://secs-simulator.pages.dev/>

#### **PLCSimulation\_Service - Scriptable Hardware Emulator (C#, WPF, JSON Scripting) | 2025.01 – 2025.06**

*Developed a high-flexibility simulation service to decouple software validation from physical PLC hardware.*

- Dynamic Behavior Modeling: Designed a script-driven architecture allowing users to load external scripts (e.g., JSON) to define varying PLC responses, error states, and timing sequences without recompiling the core application.
- Provided automated verification flow, process flow simulation, and result comparison tools
- Impact: Served as the primary validation tool for the EFEM product line, reducing on-site debugging time by 33% and enabling CI/CD integration for hardware-dependent software.
- Online: [https://github.com/KaiJuY/PLCSimulation\\_Service](https://github.com/KaiJuY/PLCSimulation_Service)

#### **Replacer – Folder Comparison & Reporting Tool (Python) 2024.11 ~ 2025.01**

*Created to solve version mismatch issues during on-site software deployment.*

- Built a high-speed directory comparison and synchronization tool using Python, featuring auto-generated change reports and differential updates.
- Designed to streamline engineering workflows where frequent build updates are critical, minimizing human error in file management.
- Relevance: Demonstrates capability in Design Flow Automation
- Online: <https://replacer.pages.dev/>

#### **AI-Based Steel Surface Defect Detection (Python, PyTorch, YOLOv11) | 2024.06 ~ 2025.06**

- Architected a customized object detection model integrating AFF, DCS, and AIE modules, improving mAP@50 to 0.836 (4% increase over baseline).
- Conducted rigorous ablation studies to validate system robustness across multiple defect types.
- Relevance: Demonstrates capability in Data Analysis, Algorithms, and Defect Inspection.
- Demo / Model Report: <https://drive.google.com/file/d/1g05Dfr809aEj-RqyROE5AAAdIA0eB6QAk/view>

#### **Embedded System Development | 2022.09 ~ 2023.01**

- Developed STM32-based embedded system using C programming for multi-module GPIO communication
- Implemented driver-level code for LCDs, Keypads, and 7-Segment displays, demonstrating solid understanding of Hardware-Software Integration and low-level resource management.

#### **Embedded Linux EFEM Controller Development | 2022.01 ~ 2022.07**

- Implemented PLC communication protocols with Ethernet for real-time equipment control and system validation
- Designed decoupled frontend-backend system using UDP-based message queue for status updates and command exchange

#### **Material Manager – Order & Inventory Management System (React, Python/Django, SQL) 2025.07 ~ 2025.09**

- Designed a full-stack web system enabling material tracking, order management, and auto consumption
- Implemented role-based operation logic, inventory deduction, and RESTful backend APIs
- Deployed frontend/backend on Render for real-time usage and open testing
- Online: <https://materialmanager-frontend.onrender.com/>

#### **Divine Whisper – Fortune Draw & AI Interpretation Web App (React, Python/FastAPI, LLM, RAG) 2025.09 ~ Present**

- Built a SaaS-style web platform with user accounts, random draw interaction, and real oracle indexing
- Integrated paid AI-based interpretation service and prompt-logic pipeline
- Designed scalable FastAPI backend with database support for draw logs & user data
- Online: <https://divine-whispers-frontend.onrender.com/>